Sri Lankan Currency Note Recognition

Individual Image Processing Project/KIC-HNDCSAI-252F-033

Project Files

Google Drive Link:

https://drive.google.com/drive/folders/1qWsOKO_OHw_9UbW823PW1cpFTLlfHFSH?usp=drive_link

Introduction and Objective

For my individual project, I chose Sri Lankan currency note recognition. The main goal is to identify the value of a note from an image. The system checks six note values: Rs. 20, Rs. 50, Rs. 100, Rs. 500, Rs. 1000, and Rs. 5000.

Methodology

The dataset is arranged into train, validation, and test folders. Each note value has its own folder. In this project there are 213 training images, 18 validation images, and 14 test images. I used OpenCV for image processing and scikit-learn for the classification part.

Algorithm Choice and Parameters

First, each image is resized to 128 x 128 pixels. Then the image is converted to grayscale and enhanced using CLAHE. This helps to improve the contrast of the note. Canny edge detection is used to find strong edges, and contours are used to show the note shape. For feature extraction, I used colour histograms from RGB, HSV, and LAB images, plus edge density and simple texture values. A Random Forest classifier with 500 trees is used for prediction.

Results

Output	File	What it shows
Input samples	dataset_samples.png	Example notes from each denomination.
Processing steps	preprocessing_steps.png	Original, enhanced, edge, and contour views.
Accuracy graph	accuracy_graph.png	Training, validation, and testing accuracy.
Confusion matrix	confusion_matrix.png	Correct and incorrect predictions by class.
Prediction example	prediction_result.png	One note image with predicted value.

Figure 1. Main image processing steps.

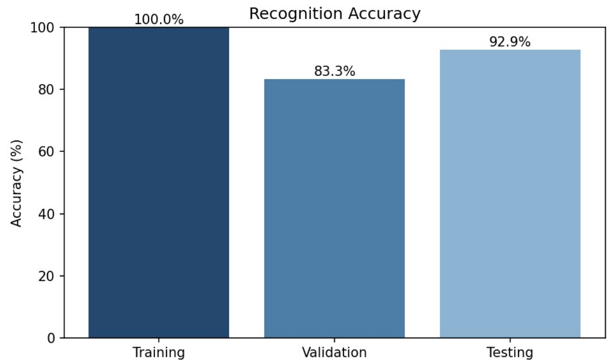

Figure 2. Accuracy summary.

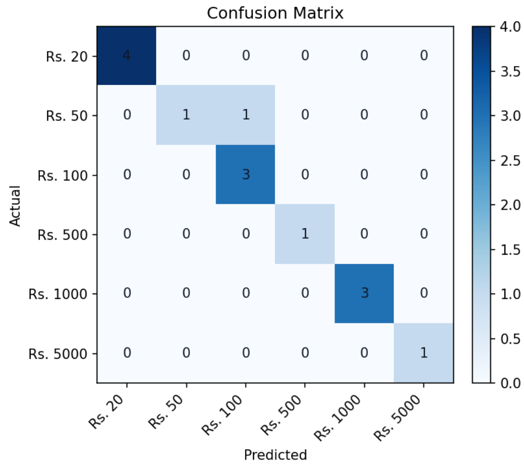

Figure 3. Confusion matrix.

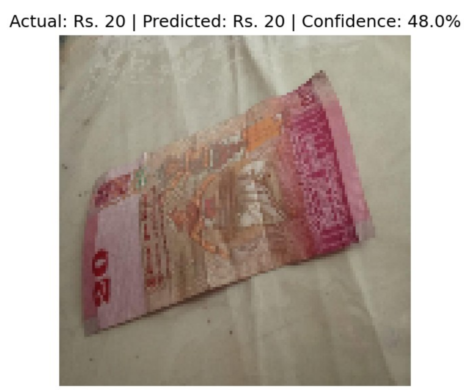

Figure 4. Example prediction.

Evaluation

Training Accuracy: 100.00%; Validation Accuracy: 83.33%; Testing Accuracy: 92.86%

The final test accuracy is 92.86%. The confusion matrix and classification report are saved in the output_images folder. Because the test set is small, the result can change a lot if only one image is predicted wrong.

	precision	recall	f1-score	support
20	1.00	1.00	1.00	4
50	1.00	0.50	0.67	2
100	0.75	1.00	0.86	3
500	1.00	1.00	1.00	1
1000	1.00	1.00	1.00	3
5000	1.00	1.00	1.00	1
accuracy			0.93	14

Discussion

This project includes the main image processing steps required for the assignment. It shows preprocessing, image enhancement, edge detection, contour detection, feature extraction, prediction, and evaluation. The output images help to compare the original note image with the processed versions.

Limitations

The main limitation is the small number of test images. There are only 14 test images, so the result may not show how the system performs on all real-world note images. This project also does not detect fake notes, because the dataset only has note value labels and not genuine/fake labels.

Conclusion

In conclusion, the project can recognize Sri Lankan currency note denominations using image processing and machine learning. The notebook includes the full code, input images, processed output images, prediction result, accuracy graph, confusion matrix, and classification report. The testing accuracy is above 75%, so the system gives a good result for this dataset.